

Joan Njoroge

Full-Stack
Web
Developer | AI
& Automation
Engineer

Portfolio: <https://kafura0.github.io/>

Github: <https://github.com/kafura0>

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Summary

Computer Science & AI graduate and **Software/Automation Engineer** with hands-on experience building full-stack applications, AI-powered tools, data-driven systems and workflow automation. Experienced in **Python, JavaScript/TypeScript, React/Next.js, APIs, databases, GIS and remote-sensing technologies**, with a focus on turning complex data and business problems into practical software solutions. Built and deployed projects including **CropSight**, a geospatial agricultural intelligence platform combining satellite data, GIS and environmental analysis. Strong interest in **AI engineering, intelligent automation, geospatial technology and scalable product development**, with a proven ability to independently take ideas from concept through implementation and deployment.

Core Competencies

Full-Stack Development • API Integration • GIS and Remote Sensing Engineer • Enterprise Systems • Prompt Engineering • Solution Architecture • Cloud Deployment • Stakeholder Collaboration • AI Engineering • Workflow Automation •

Experience

JOAT Kenya / AI-assisted automation engineer

Feb, 2025 - Apr, 2026, Kiambu Rd, Kenya

- Led the design and implementation of AI-powered automation solutions that streamlined business workflows and reduced manual effort.
- Architected scalable full-stack applications integrating AI services, APIs, cloud databases, and workflow automation platforms.
- Partnered with non-technical stakeholders and designers to build and maintain web applications.
- Built reusable automation pipelines using n8n, REST APIs, and LLMs to improve operational efficiency.
- Evaluated emerging AI technologies and selected practical solutions based on scalability, maintainability, and business value.
- Designed modular software architectures supporting multi-tenant SaaS applications.

SmartBrains Kenya / Web development and robotics teacher-EduTechAI

May, 2024- Aug, 2025, Nairobi

- Teach a range of programming languages (HTML/CSS, JavaScript, Arduino) and software development practices. Evaluate student progress and provide helpful feedback to facilitate skill improvement.
- Organize coding projects, competitions, and workshops to promote hands-on learning.

Education

University of Nottingham / Bachelor of Science – Computer Science & Artificial Intelligence - Sept 2016 - May 2021, Nottingham, United Kingdom

Computer-Science-with-Artificial-Intelligence-BSc <https://www.nottingham.ac.uk/ugstudy/course/>

Moringa school / FlatIron School Data Science Course - Apr, 2023 - Apr 2024

Understanding Data Science and AI concepts and algorithms.

Technical Skills

Frontend: React, Next.js, Tailwind CSS | DevOps: Docker, Git, GitHub, Render, Vercel | Backend: FastAPI, Node.js, PostgreSQL, Supabase, REST APIs | Languages: Python, JavaScript, TypeScript, HTML, CSS | AI: Claude, Gemini, OpenAI, OpenRouter, Prompt Engineering, LLM Integration, AI Workflow Design | Automation: n8n, Selenium, REST APIs, Webhooks, Workflow Orchestration

Projects

CropSight – Geospatial Agricultural Intelligence Platform | 2026

GitHub: https://github.com/kafura0/gis_AI/

Prototype: <https://cropsight-72co.onrender.com/>

- Built and deployed a full-stack GIS platform combining **satellite imagery, remote sensing, environmental data and crop suitability modelling** for agricultural decision support.
- Implemented interactive point/polygon analysis, field statistics, vegetation monitoring and spectral indices including **NDVI, EVI, SAVI and NDWI**.
- Developed multi-factor crop suitability scoring and integrated geospatial APIs to transform satellite and environmental data into actionable agricultural insights.
- Designed the architecture for future **AI-assisted analysis, automated reporting and scalable geospatial decision-support workflows**.

Tech: Python · GIS · Remote Sensing · React/Next.js · REST APIs · PostgreSQL · Satellite Data · Data Visualisation

AI content creation engine Nov, 2025 - Present

GitHub: https://github.com/kafura0/ai_content_engine

- Built an AI-assisted content workflow system powered by Claude and Gemini models through OpenRouter.
- Structured prompt workflows for generating, refining and organizing social media and content outputs.
- Focused on improving output quality and workflow consistency while optimizing lower-cost AI model usage.
- Explored automation-oriented publishing and content management workflows.

BI Platform – AI-Powered Business Intelligence SaaS Feb, 2026 - Present

- Designed and built a production-grade multi-tenant Business Intelligence platform enabling organizations to upload operational data and generate AI-powered dashboards.
- Implemented natural language querying using LLMs to simplify business reporting.
- Designed the application for scalability using cloud-native technologies.
- Focused on reducing manual reporting while improving access to operational insights.

